ELLIPTIC CURVES

In this section, we begin our study of elliptic curves defined over fields. There are numerous ways in which one can look at elliptic curves, and it is imperative that you study these fascinating objects as much by yourselves as will be studied in the class.

1. Elliptic curves

Elliptic curves are essentially smooth projective curves of genus 1. Recall that this means that there exists a (unique up to scalar) nonzero holomorphic differential on such curves. What is more, it turns out that such curves can be given the structure of an abelian group. Of course, a requirement for this is the need to fix a special element which will fulfill the role of the identity element of the group.

Definition 1.1. An *elliptic curve* is a pair (E, O) where E is a smooth projective curve of genus one and $O \in E$. We will simply say 'E is an elliptic curve'; the point O is tacitly understood. E is defined over K, if E is defined over K as a projective curve, and $O \in E(K)$.

The Riemann-Roch theorem tells us that elliptic curves are the same as smooth cubic curves in a plane. The latter are also called 'smooth Weierstrass plane cubic curves'.

Theorem 1.2. *LEt E be an elliptic curve defined over K.*

(i) There exist functions $x, y \in K(E)$ such that the map

$$\phi : E \to \mathbb{P}^2, \quad \phi = [x : y : 1]$$

gives an isomorphism of E/K onto a curve given by the following Weierstrass *equation*

$$C : Y^2 + a_1XY + a_3Y = X^3 + a_2X^2 + a_4X + a_6$$

with coefficients $a_1, \ldots, a_6$ in K such that $\phi(O) = [0 : 1 : 0]$. The functions x, y are called Weierstrass coordinates for E.

(ii) Any two Weierstrass equations for E are related by a change of variables of the form

$$X = u^2X' + r, Y = u^3Y' + su^2X' + t$$

where $u \in K^*$ and $r, s, t \in K$.

(iii) Conversely, every smooth cubic curve C given by a Weierstrass equation is an elliptic curve defined over K with base point $O = [0 : 1 : 0]$.

Proof. For every integer $n \geq 1$, Riemann-Roch Theorem gives

$$\dim_K L(n(O)) = n.$$

Choose $x \in L(2(O)) \setminus L((O))$ so that $\{1, x\}$ is a basis for $L(2(O))$, and then choose $y \in L(3(O)) \setminus L(2(O))$ so that $\{1, x, y\}$ is a basis for $L(3(O))$. It follows that

$$\mathrm{ord}_O(x) = -2, \quad \mathrm{ord}_O(y) = -3,$$

and neither x nor y has a pole away from O.

The vector space $L(6(O))$ has dimension 6, but it contains, in particular, the seven functions

$$1,\ x,\ y,\ x^2,\ xy,\ y^2,\ x^3.$$

Hence, they must be linearly dependent over K. The coefficients of y^2 and x^3 in such a relation must both be nonzero by a comparison of orders of poles at O. After rescaling x and y and moving the remaining terms to the appropriate sides, the resulting relation has the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Thus the rational map

$$\phi = [x : y : 1] : E \longrightarrow \mathbb{P}^2$$

has image contained in a Weierstrass cubic C. Since E is smooth, it is a morphism, and the pole orders show that $\phi(O) = [0:1:0]$.

The map $[x:1] : E \to \mathbb{P}^1$ has degree 2, since x has a unique pole of order 2, and similarly $[y:1]$ has degree 3. Hence $[K(E):K(x)] = 2, [K(E):K(y)] = 3$. The degree $[K(E):K(x,y)]$ thus divides both 2 and 3, and therefore $K(E) = K(x,y)$. It follows that ϕ has degree 1. Note that this still does not force ϕ to be an isomorphism since we don't a priori know C to be smooth. But generalities from algebraic geometry of normalizations of varieties allow us to deduce that C in fact must be smooth if it admits a degree one morphism from a smooth curve E, and hence ϕ must be an isomorphism.

For (ii), suppose x, y and x', y' are two pairs of Weierstrass coordinates. Since $\{1, x\}$ and $\{1, x'\}$ are both bases for $L(2(O))$, and $\{1, x, y\}$ and $\{1, x', y'\}$ are both bases for $L(3(O))$, comparison of these bases gives $x = u_1 x' + r, y = u_2 y' + s_2 x' + t$ for suitable constants. Comparing the coefficients gives the stated form.

For (iii), the projective closure of a smooth Weierstrass equation is a smooth plane cubic with the K-rational point $[0:1:0]$. It is a standard fact that a smooth plane cubic has genus one. Equivalently, one may check that the differential

$$\omega = \frac{dx}{2y + a_1 x + a_3}$$

has neither zeros nor poles, and hence its canonical divisor has degree zero. Thus the curve, together with the point $[0:1:0]$, is an elliptic curve over K. □

Remark 1.3. The projective equation corresponding to the affine Weierstrass equation is

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3.$$

Its only point on the line $Z = 0$ is $O = [0:1:0]$, and the line at infinity meets the cubic at O with multiplicity 3.

1.1. **The group law from divisor classes.** We now use the Riemann–Roch theorem to construct the group law intrinsically. The following elementary fact is the source of the uniqueness in the construction.

Lemma 1.4. *Let C be a curve of genus one and let $P, Q \in C$. Then*

$$(P) \sim (Q)$$

if and only if $P = Q$.

Proof. Suppose $(P) \sim (Q)$. Then there is a rational function f satisfying $\operatorname{div}(f) = (P) - (Q)$. Thus $f \in L((Q))$. But Riemann-Roch gives $\dim L((Q)) = 1$, and this space already contains the constant functions. Hence f is constant, so its divisor is zero and $P = Q$. The converse is immediate. □

Proposition 1.5. *Let (E, O) be an elliptic curve.*

(i) For every divisor $D \in \operatorname{Div}^0(E)$ there is a unique point $P \in E$ such that

$$D \sim (P) - (O).$$

(ii) The assignment $\sigma(D) = P$ for P as above gives a bijection

$$\sigma : \operatorname{Pic}^0(E) \longrightarrow E.$$

Its inverse is

$$\kappa : E \longrightarrow \operatorname{Pic}^0(E), \qquad P \longmapsto [(P) - (O)].$$

(iii) Transporting the abelian group structure on $\operatorname{Pic}^0(E)$ through σ gives an abelian group law on E whose identity is O.

Proof. Let $D \in \operatorname{Div}^0(E)$. Since E has genus one and $\deg(D + (O)) = 1$, Riemann-Roch gives $\dim L(D + (O)) = 1$. Choose a nonzero $f \in L(D + (O))$. Then

$$\operatorname{div}(f) + D + (O)$$

is an effective divisor of degree 1. Hence it must be equal to (P) for some point $P \in E$. Thus $D \sim (P) - (O)$.

The uniqueness of P follows from the preceding lemma. It follows immediately that two degree-zero divisors have the same image under σ precisely when they are linearly equivalent, so σ descends to the stated bijection. The remaining assertion follows by transporting addition from $\operatorname{Pic}^0(E)$ to E. □

Thus the sum of two points P and Q on E, say $P+Q=R$ is characterised by $(R)-(O)\sim(P)-(O)+(Q)-(O)$. Equivalently,

$$(P)+(Q)\sim(P+Q)+(O). \tag{1.1}$$

The inverse of a point P is the unique point $-P$ satisfying

$$(P)+(-P)\sim 2(O).$$

Remark 1.6. If E is defined over K, then the construction is compatible with the action of G_K. Indeed, uniqueness assertion above gives $(P+Q)^\tau = P^\tau + Q^\tau$ for every $\tau\in G_K$. Hence $E(K)$ is an abelian subgroup of $E(\bar{K})$.

Corollary 1.7. *Let $D=\sum_{P\in E} n_P P\in \mathrm{Div}(E)$. Then D is principal if and only if*

$$\sum_{P\in E} n_P = 0 \text{ and } \sum_{P\in E}[n_P]P = O,$$

where the second sum is now taken using the group law on E. In particular, $[n_P]P$ refers to the point obtained by adding the point P to itself n_P many times.

Proof. The first condition says that D has degree zero. For a degree-zero divisor, the proposition gives

$$D\sim 0 \iff \sigma(D)=O.$$

Since σ is additive after passing to divisor classes by construction of the abelian group structure over E, $\sigma(D)=\sum_{P\in E}[n_P]P$.

This proves the assertion. □